

MOHIT TRIPATHI

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PROFESSIONAL SUMMARY

Senior Embedded Systems, IoT, Embedded AI and Robotics Software Engineer with 9+ years of experience building production-grade connected products, Linux/RTOS firmware, edge gateways and cloud-integrated device platforms. Strong hands-on background in C/C++, Python, Go, embedded Linux, FreeRTOS, device protocols, sensor integration, OTA/FOTA, Industrial IoT, technical architecture and production debugging. Robotics and Physical AI project work includes **ROS 2**, **Nav2**, **SLAM**, **AMCL**, **MoveIt 2**, **Gazebo/RViz**, computer vision and reinforcement-learning-based control. Experienced in leading cross-functional delivery across software, hardware, cloud, validation, OEM and product teams.

CORE SKILLS

Embedded & Linux	C, C++, Python, Go, embedded Linux/Ubuntu, RTOS/FreeRTOS, multithreading, IPC, shell scripting, watchdogs, fault recovery, board-level debugging and performance analysis
Hardware & Protocols	UART, I2C, SPI, Modbus, JTAG, D-Bus, MQTT, TCP/IP, HTTP/HTTPS, REST/JSON; ESP32/ESP8266, TI CC3200, Raspberry Pi, L506, NXP i.MX8M Plus, PLCs, sensors and edge gateways
Robotics & Physical AI	ROS 2 nodes/topics/services/actions, QoS, lifecycle nodes, TF2, URDF/Xacro, Nav2, SLAM Toolbox, AMCL, MoveIt 2, Gazebo, RViz, Webots, odometry, behaviour trees and robot-control fundamentals
Embedded AI, Vision & Edge AI	Artificial intelligence, machine learning, reinforcement learning and LLM-based automation; OpenCV, YOLO-based detection, multi-object tracking, PyTorch/TensorFlow familiarity, ONNX, TensorRT/TFLite deployment workflows, quantisation and latency/resource analysis
Cloud & Deployment	Azure IoT Edge, Event Hubs, Function Apps, Cosmos DB, SQL, Application Insights, AWS IoT, PubNub, Docker, telemetry pipelines, secure OTA/FOTA, monitoring and scalable device operations
Engineering Leadership	System architecture, interface definition, requirements translation, design/code reviews, mentoring, unit/HIL testing, product validation, stakeholder management and OEM/vendor coordination; Git, GitHub, Bitbucket, Cursor and GitHub Copilot

PROFESSIONAL EXPERIENCE

Maersk Global Service Centre

Oct 2023 - Jun 2026

Software Engineer, Edge IoT & Vision AI

- Built Python-based edge workflows for UART capture, **Carrier ML3** and **Daikin ML5** protocol decoding, sensor-data validation, restructuring and secure cloud transmission across 40,000+ refrigerated-container IoT devices.
- Integrated Azure Event Hubs and Function Apps for real-time telemetry processing; optimised ingestion and batching to improve throughput and reduce CPU/RAM cost.
- Applied LLM-based automation using Cursor and GitHub Copilot for prompt-assisted code generation, parser refactoring, unit-test creation and technical documentation, with human review before production release.
- Strong background in networking concepts including TCP, TLS, HTTPS, Bluetooth and Wi-Fi for secure edge connectivity.
- Improved structured log accuracy by 70%, raising usable output quality to approximately 90% through rollover handling, parser fixes and corrupted/incomplete-data validation.
- Developed a container-detection and tracking PoC using YOLO-based computer vision and multi-object tracking for real-time counting, identity persistence and edge monitoring.
- Implemented and tested FOTA, observability, health monitoring, communication recovery and production diagnostics; supported Cosmos DB, SQL and Application Insights workflows.
- Onboarded Carrier and Daikin refrigerated-container integrations, expanding multi-OEM support while coordinating interfaces across edge, cloud, database and validation teams.

Panasonic Life Solutions India Pvt. Ltd. (IIC)

Dec 2020 - Oct 2023

Tech Lead, IoT Engineering

- Led architecture and delivery of the **Panasonic Light Management System (LMS)** on the **MirAie Spatial Solution (MSS)** platform, translating product requirements into gateway, firmware, cloud-interface and validation plans.
- Developed Go-based Linux gateway services using MQTT, HTTPS, REST/JSON and D-Bus for **DALI** lighting control, PIR/ALS/temperature/humidity sensors, scenes/groups, log upload and FOTA.
- Led firmware development for **Anchor Captor i-Kraft Smart Fan**, **Anchor Smart Retrofit Controllers**, **Anchor Roma Switches (5 models)** and smart plugs using ESP8266 and Espressif RTOS SDK, integrated with the **Panasonic MirAie App**; contributed the **Panasonic AC PM2.5** algorithm and **BL0937/HLW8032** energy-measurement logic.
- Optimised gateway services for CPU usage, response time, watchdog behaviour, communication recovery and field diagnostics.
- Led design/code reviews, mentoring, OEM/vendor coordination, line testing, validation and mass-production readiness across firmware, gateway, application and hardware teams.
- Supported Roma setup and technical coordination across teams in Singapore, Thailand, Japan, Saudi Arabia and India.

Truminds Software Systems

Jul 2020 - Dec 2020

Senior Software Engineer

- Developed Anchor Roma smart-panel models (5 variants) using ESP8266 and Espressif RTOS SDK and integrated them with the **Panasonic MirAIe App**.
- Coordinated daily with stakeholders and cross-functional teams while performing testing, debugging and release support.

BlickX Technologies Pvt. Ltd.

Jan 2020 - May 2020

Senior Embedded Engineer

- Enhanced and deployed an Industrial IoT 4.0 product-monitoring system using **Delta PLC**, Raspberry Pi, Python/Flask, SQL/SQLite and FreeRTOS.
- Implemented device synchronisation, REST API integrations, SQLite queries/triggers and client-driven debugging for reliable production operation.

ThingsCloud Technologies Pvt. Ltd.

Mar 2018 - Jan 2020

Embedded Engineer

- Developed an **L506**-based 4G IoT gateway on Linux using C and the **ThingsWiFi** product on Windows using C; connected **TI CC3200**, NodeMCU and **Raspberry Pi 3** to **AWS IoT** and **PubNub** using MQTT, TCP/IP, REST and JSON.
- Integrated **ThingsCloud Solar IoT** devices with **Delta**, **Zever** and **Growatt** solar inverters using Modbus and transmitted inverter telemetry to cloud services.
- Implemented Raspberry Pi socket communication with HTTP servers, SD-card data-reading tools, OTA, RTC and TI-RTOS-based device features.
- Developed Qt-based motor-controller utilities and BLE monitoring for ambient temperature and energy usage; supported testing, debugging and mass production.
- Implemented HTTPS server communication on **TI CC3200** using Embedded C.

Zorro Fox Technologies Pvt. Ltd.

Jun 2015 - Aug 2016

Research Engineer

- Developed and mentored embedded, MATLAB, IoT and robotics projects, including Linux/Embedded C home automation, Python weather monitoring and smart street-light control.
- Built Atmega16-based DTMF automation, parking-indicator and bomb/fire-detection robot prototypes; delivered technical training and demonstrations.

SELECTED ROBOTICS, PHYSICAL AI & EMBEDDED AI PROJECTS

ROS 2 Autonomous Navigation and Manipulation System

- Built a modular **ROS 2** simulation integrating differential-drive control, odometry, LiDAR, RGB camera, TF2, lifecycle nodes and operator interfaces.
- Implemented **SLAM Toolbox** mapping, **AMCL** localisation, **Nav2** planning, layered costmaps, obstacle avoidance, controller tuning, recovery behaviours and behaviour-tree navigation.
- Configured **MoveIt 2** workflows for a 6-DOF manipulator covering URDF/Xacro, planning groups, inverse kinematics, collision-aware planning and trajectory execution; debugged QoS, transforms and simulation issues in **Gazebo/RViz**.

Learning-Based Quadruped Locomotion

- Implemented an **NVIDIA Isaac Gym**-style reinforcement-learning workflow inspired by DreamWaQ, covering reward design for velocity tracking, stability, balance, energy efficiency and disturbance recovery.
- Applied domain randomisation and sim-to-real principles across terrain and system variations; familiar with **NVIDIA Isaac Sim** concepts for robotics simulation and synthetic workflows.

Edge Vision and Embedded AI Deployment Pipeline

- Designed a camera pipeline for preprocessing, inference, tracking, counting, event generation and monitoring, with planned **ROS 2** integration.
- Evaluated ONNX, TensorRT, TFLite and quantisation workflows for low-latency inference, including frame-loss, accuracy, latency and compute-utilisation monitoring.

EDUCATION

Indian Institute of Science (IISc): Advanced Certification Programme in NextGen Robotics with AI | Robotics, AI/ML, reinforcement learning, computer vision, **ROS 2**, **Gazebo**, **Nav2**, **MoveIt 2**, Webots and Edge AI

Staffordshire University: B.E. (Hons.) Mechatronics, 2015 | Final-year project: Pump Discharge Flow Variation with VFD and PLC

RESEARCH & RECOGNITION

Research: "Pump Discharge Flow Variation with VFD and PLC" published at the ICACCT IEEE Conference

Recognition: Super Star Award for successful launch of Anchor Smart Fan at Panasonic India

ADDITIONAL INFORMATION

Certification: Data Analysis with Python (Pandas, NumPy, Matplotlib and exploratory analysis) | **Languages:** English, Hindi